

Supplementary Material

KNMTG: A Korean Nationwide Multimodal Transit Graph for Time-Dependent Midpoint Search

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S1. Transfer Penalty Parameters

Table 1 summarises all inter-mode transfer penalties implemented in the routing engine. Penalties were calibrated empirically to ensure that bus-mediated shortcuts do not spuriously undercut rail boarding costs.

Table 1: Transfer penalty values by transition type

From mode	To mode	Penalty (min)
Any	KTX/SRT (boarding)	20
Any	GTX-A (boarding)	8
Any	Intercity bus (boarding)	15
Any	BRT (boarding)	5
Subway	Subway (same line)	0
Subway	Subway (different line)	5
KTX/SRT	Subway (at co-located station)	10

S2. Complete Routing Test Suite

All 33 routing test cases used for system validation are reported below (Table 2). Column “KNMTG” shows computed travel time; “Expected” shows the range defined in the test specification; a checkmark indicates the result falls within the range.

Table 2: Complete routing test results (33 O-D pairs)

Origin	Destination	KNMTG (min)	Expected (min)	Note
Gangnam	Yeoksamm	2.3	1–6	Adjacent, Line 2
Jongak	Sijeong	2.7	1–6	Line 1 adjacent
Jamsil	Jamsil-Saenae	2.3	1–6	Line 2 adjacent
Gangnam	Gyodae	2.3	1–8	Line 2, 2 stops
Gangnam	Sadang	9.2	8–22	Line 2, 7 stops
Jongno-3-ga	Jamsil	34.9	18–40	Line 2 + Line 5 transfer
Gangnam	Hongik Univ.	39.1	25–45	Line 2, 12 stops
Gangnam	Seoul Station	31.4	25–45	Line 2 + Line 1 transfer
Jamsil	Gangnam	13.8	8–22	Line 2, 7 stops
Gangnam	Hyehwa	39.7	28–50	1–2 transfers
Gangnam	Gwanghwamun	40.0	28–50	Line 2 + Line 5 transfer
Gangnam	Gongdeok	35.0	25–45	Multiple paths
Gangnam	Jeongja	22.5	18–35	Sinbundang 5 stops direct
Gangnam	Pangyo	18.0	14–30	Sinbundang 4 stops direct
Seoul Stn.	Suwon	22.3	20–80	Saemaul direct
Incheon	Seoul Stn.	72.4	60–95	Line 1 end-to-end
Incheon	Sindorim	56.7	45–80	Line 1
Incheon	Mullae	62.5	50–90	Line 1 + Line 2
Incheon	Gangnam	60.0	55–115	Express bus direct
Yongmun	Mullae	83.5	80–130	Gyeonggi Yangpyeong
Yongmun	Wangsimni	57.0	55–95	Gyeongui-Jungang direct
Yongmun	Cheongnyangni	39.0	35–90	Gyeongui-Jungang
Danggogae	Sadang	62.4	50–80	Line 4 end-to-end
Banghwa	Hanam-Geomdansan	100.6	75–115	Line 5 Sangil branch end
Banghwa	Macheon	95.7	70–100	Line 5 Macheon branch end
Seoul KTX	Busan KTX	141.0	140–200	KTX direct
Seoul KTX	Dongdaegu KTX	101.0	95–140	KTX direct
Seoul KTX	Gwangju-Songjeong KTX	110.0	100–145	Honam KTX
Seoul KTX	Gangneung KTX	106.0	75–130	Gangneung KTX
Seoul KTX	Daejeon KTX	56.0	50–85	KTX direct
Gangnam	Busan KTX	156.0	150–220	Suseo (SRT) or Seoul KTX

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Origin	Destination	KNMTG (min)	Expected (min)	Note
Gangnam	Dongdaegu KTX	117.9	110–180	Suseo (SRT) or Seoul KTX
Busan KTX	Gwangju-Songjeong KTX	162.0	150–280	Via Osong transfer

S3. Degree Distribution Data

The full degree distribution of KNMTG v1.1 is shown in Figure 1. Degree 2 (chain-like subway segments) accounts for 51.3% of all nodes. The long tail to degree 41 (Express Bus Terminal, Seoul) reflects the multimodal hub function of major intercity terminals.

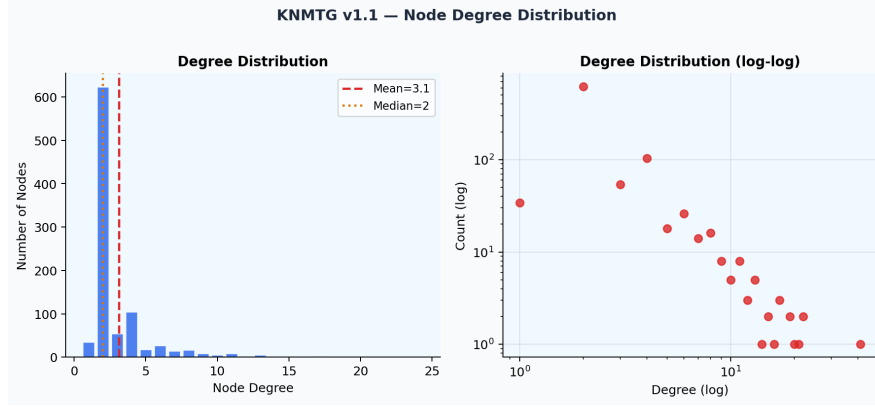


Figure 1: Degree distribution of KNMTG v1.1 (932 nodes).

S4. Midpoint Scoring Function

Given n origins and travel-time maps $d_i : V \rightarrow \mathbb{R}_{\geq 0}$, the composite score for candidate node v under strategy s is:

$$\text{score}_s(v) = w_{\max}^s \cdot \max_i d_i(v) + w_{\text{sum}}^s \cdot \sum_i d_i(v) + w_{\sigma}^s \cdot \sigma(d_i(v)) - w_{\text{pr}}^s \cdot \text{PageRank}(v) \quad (1)$$

Table 3 lists the weight vectors for each strategy.

Table 3: Strategy weight parameters				
Strategy	$w_{\max}$	w_{sum}	w_{σ}	w_{pr}
Balanced (default)	0.5	0.3	0.2	50
Minimize-max	1.0	0.1	0.1	20
Minimize-sum	0.1	1.0	0.1	20

S5. Data Availability

The full KNMTG v1.1 dataset (nodes, edges, routing code) is publicly available at <https://github.com/yjkim0123/knmtg> under CC-BY 4.0 (data) and MIT (code) licenses.